

PERSPECTIVE



Clinical Cardiology at the Crossroads: Perspectives on the Next Era at *Circulation*

Jennifer E. Ho , MD; Ryan J. Tedford , MD

*Where is the wisdom we have lost in knowledge?
Where is the knowledge we have lost in information?*

—T.S. Eliot

For decades, *Circulation* has fundamentally shaped how we care for patients with cardiovascular disease, from the first report of external defibrillation in humans,¹ the initial identification of cardiovascular risk factors² in the Framingham Heart Study, to the landmark studies that have broadly shaped modern cardiology practice across prevention, imaging, intervention, electrophysiology, and heart failure. The enduring influence of *Circulation* has stemmed not only from publishing important discoveries, but from bringing together mechanistic understanding, clinical investigation, and patient-centered impact. As we build upon this deep legacy and consider the next decade, it is important to note that today, as clinicians, we are confronted with an unprecedented scale and velocity of cardiovascular information. Indeed, the central challenge facing clinicians is no longer the acquisition of data, but rather the transformation and integration of data into knowledge, and knowledge into clinical acumen and wisdom.

As we look forward, we envision that the focus of clinical cardiology at *Circulation* will remain centered on clinical impact, a “north star” that has guided the journal throughout. However, our goal as deputy editors is to evolve with this unparalleled expansion of data, to curate, distill, and contextualize the most impactful work for the clinicians and patients of today. We believe that the role of a leading journal is no longer simply to publish information, but to identify the studies that most rigorously and meaningfully change how clinicians understand disease and care for patients. As we endeavor to seek out the most impactful

work, there are 3 guiding questions to anchor us: Is the research done well? Is it novel? Is it clinically actionable? In our view, the best clinical cardiology papers should pass the “coffee table test”: clinicians should feel compelled to stop, read, discuss, and reconsider practice.

In this context, what should be our priorities? Clinical cardiology occupies a unique position within cardiovascular science, serving as the nexus in which mechanistic discovery, technological innovation, clinical investigation, and population health converge (Figure). The most influential studies will not simply generate more data; they will reveal new biological paradigms, refine our understanding of disease, and provide actionable insights that improve patient outcomes. We envision 3 broad areas of emphasis and scientific discovery: (1) mechanisms of disease, which includes translational human investigations that advance understanding of cardiovascular physiology, hemodynamics, and pathobiology, leveraging deep phenotyping through imaging, molecular profiling, genomics, proteomics, and other emerging technologies to uncover clinically relevant mechanisms; (2) therapeutic innovation, which includes studies that evaluate and accelerate the adoption of effective therapies, including clinical and pragmatic trials, investigations of drugs and devices, comparative effectiveness research, and implementation science designed to bridge the gap between discovery and practice; and (3) systems and population impact; we seek research that improves cardiovascular health at scale through advances in care delivery, health policy, prevention, and population science, with particular emphasis on globally relevant questions, diverse populations, and scalable solutions that can be implemented across a broad range of healthcare environments. Together, these domains reflect our central

Key Words: cardiology ■ cardiovascular diseases

The opinions expressed in this article are not necessarily those of the editors or of the American Heart Association.

Correspondence to: Jennifer E. Ho, MD, Harvard Medical School, Division of Cardiology, Beth Israel Deaconess Medical Center, 330 Brookline Ave, E/CLS 945, Boston, MA 02215-5491, Email jho@bidmc.harvard.edu; and Ryan J. Tedford, MD, Medical University of South Carolina, 30 Courtenay Dr, BM215, MSC592, Charleston, SC 29425, Email tedfordr@musc.edu

The podcast and transcript are available as Supplemental Material at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCULATIONAHA.126.080205>.

© 2026 American Heart Association, Inc.

Circulation is available at www.ahajournals.org/journal/circ

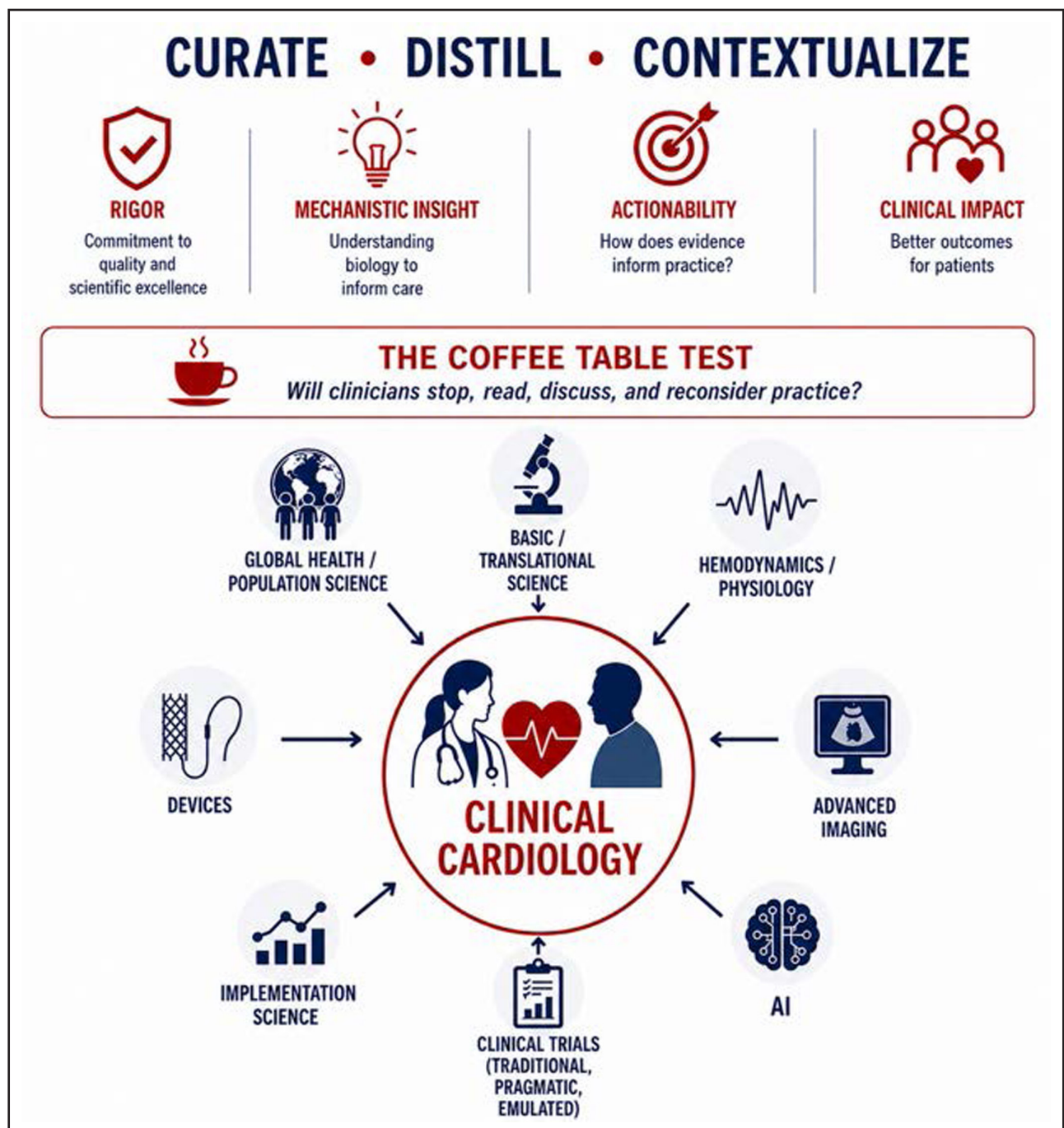


Figure. Clinical cardiology at the crossroads.

Clinical cardiology is uniquely positioned at the convergence of research studies spanning mechanisms of disease, therapeutic innovation, and systems and population impact. The figure was developed and edited using OpenAI, 2026, ChatGPT (May 31 version). AI indicates artificial intelligence.

goal: to publish science that not only advances knowledge, but also meaningfully changes the way cardiovascular disease is understood, prevented, and treated.

Across these areas of emphasis, several trends are poised to reshape clinical cardiovascular investigation. Advances in molecular profiling, multimodal imaging, remote monitoring, and digital health are rapidly expanding our ability to characterize cardiovascular disease at a level of granularity that was previously unimaginable. The

challenge is no longer simply generating data, but sorting “signal from noise” (ie, which biological, physiological, and clinical signals converge to meaningfully inform diagnosis, prognosis, or treatment). As these technologies mature, we anticipate a growing shift from disease classifications based primarily on clinical syndromes toward approaches that identify biologically and mechanistically distinct patient subgroups, enabling more precise prevention and therapeutic strategies.³

Artificial intelligence will undoubtedly accelerate this transformation, yet we need to assure that scientific rigor is not sacrificed with its adoption and integration. The most important advances will move beyond risk prediction and help uncover disease mechanisms, identify actionable therapeutic targets, improve clinical decision-making, streamline workflows, and expand access to high-quality cardiovascular care. Equally important, the field must maintain rigorous standards for phenotyping, transparency, validation, reproducibility, and clinical implementation.⁴ Novel algorithms that marginally improve prediction metrics yet fail to influence clinical decisions or patient outcomes are unlikely to meaningfully advance cardiovascular care.

The greatest opportunities lie at the intersection of disciplines that have traditionally evolved in parallel. Increasingly, the most informative cardiovascular investigations will integrate across molecular profiling, imaging, hemodynamics, device and wearable data, clinical trials, and real-world health care measures. We envision clinical cardiology sitting at the convergence of these research approaches and domains. The central question is not whether more data can be generated, but how disparate sources of information can be combined to yield insights that would not be apparent from any individual approach alone.⁵ In other words, the next generation of discoveries may emerge not from a single approach, but from the ability to work across disciplines to connect fundamental biological insights to clinical care.

We are honored to serve the readers, authors, and reviewers of *Circulation* during this exciting period of discovery. As the volume, complexity, and velocity of cardiovascular data continue to expand, the challenge articulated by T.S. Eliot remains as relevant today as ever: transforming information into knowledge and knowledge into wisdom. We look forward to the investigations that will define the next era of clinical cardiology and invite submissions that challenge paradigms, illuminate disease mechanisms, rigorously evaluate therapies, and improve the lives of patients worldwide. Through a commitment to scientific rigor, mechanistic insight, clinical actionability,

and patient impact, our goal is to curate, distill, and contextualize the science that most meaningfully advances cardiovascular care.

ARTICLE INFORMATION

Affiliations

Cardiovascular Institute and Division of Cardiology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA (J.E.H.). Harvard Medical School, Boston, MA (J.E.H.). Division of Cardiology, Department of Medicine, Medical University of South Carolina, Charleston, SC (R.J.T.).

Disclosures

J.E.H. has received consultant fees from Eli Lilly. R.J.T. reports general disclosures to include consulting relationships with and receiving honorarium/consulting fees from Abbott, Acorn, Adona, Aria CV Inc., Boston Scientific, CORTER, CVRx, Endotronix, Edwards LifeSciences, Fauna Bio, Gradient, Imbria, Inhibikase, Merck, Mitre, Morphic Therapeutics, Pulmovant, Restore Medical, Tempus AI, 35Pharma, and United Therapeutics. He serves on steering committees for Abbott, Edwards, Endotronix, Gradient, Merck, Restore Medical, and Tempus AI, as well as a research advisory board for Abiomed.

REFERENCES

1. Gibson W, Linenthal AJ, Norman LR, Paul MH, Zoll PM. The effect of external electric currents on the heart; control of cardiac rhythm and induction and termination of cardiac arrhythmias. *Circulation*. 1956;14:745–756. doi: 10.1161/01.cir.14.5.745
2. Wilson PW, D'Agostino RB, Levy D, Belanger AM, Silbershatz H, Kannel WB. Prediction of coronary heart disease using risk factor categories. *Circulation*. 1998;97:1837–1847. doi: 10.1161/01.cir.97.18.1837
3. Krittanawong C. Future physicians in the era of precision cardiovascular medicine. *Circulation*. 2017;136:1572–1574. doi: 10.1161/CIRCULATIONAHA.117.029676
4. Aroundas AA, Narayan SM, Arnett DK, Spector-Bagdady K, Bennett DA, Celi LA, Friedman PA, Gollob MH, Hall JL, Kwitek AE, et al; American Heart Association Institute for Precision Cardiovascular Medicine; Council on Cardiovascular and Stroke Nursing; Council on Lifelong Congenital Heart Disease and Heart Health in the Young; Council on Cardiovascular Radiology and Intervention; Council on Hypertension; Council on the Kidney in Cardiovascular Disease; and Stroke Council, Council on C, Stroke N, Council on Lifelong Congenital Heart D, Heart Health in the Y, Council on Cardiovascular R, Intervention, Council on H, Council on the Kidney in Cardiovascular D and Stroke C. Use of artificial intelligence in improving outcomes in heart disease: a scientific statement from the American Heart Association. *Circulation*. 2024;149:e1028–e1050. doi: 10.1161/CIR.0000000000001201
5. Wang RS, Maron BA, Loscalzo J. Multiomics network medicine approaches to precision medicine and therapeutics in cardiovascular diseases. *Arterioscler Thromb Vasc Biol*. 2023;43:493–503. doi: 10.1161/ATVBAHA.122.318731